

FlyRank AI Fluency — Prompt Ladder

Track

Search Ranking Improvement

Goal

Improve a prompt used to understand and frame the search-ranking ML problem, changing only one prompt layer at a time.

Version 0 — Weak Baseline

Prompt

Help me improve search ranking using machine learning.

Output — Baseline

Machine learning can improve search ranking by using relevant features and training a model to rank search results. You can collect data, engineer features, train a ranking model, and evaluate it using suitable metrics.

What Changed?

Nothing. This is the original weak prompt.

What Improved?

Nothing useful. The answer was generic and could apply to almost any search-ranking project.

What Still Failed?

It did not know my internship context, the actual problem, the data, the baseline or what I was expected to produce.

What I Would Try Next

Add a **clearer goal**.

Version 1 — Clearer Goal

Layer Added

Clearer goal

Prompt

Help me frame a machine-learning approach for improving the existing search-ranking pipeline. I want to understand the problem and decide what should be improved before choosing a complex model.

Output — Representative Excerpt

First define the ranking problem and identify the current baseline. Then inspect the available features and evaluation metric. After establishing the baseline, identify where the performance gap comes from before selecting a model.

What Changed?

I added a specific goal: **understand and frame the existing ranking problem before choosing a model**.

What Improved?

The response became more focused on **problem definition and baseline analysis** instead of giving a generic list of ML techniques.

What Still Failed?

It still did not know that this was part of my **FlyRank internship** or what I had already observed.

What I Would Try Next

Add **audience**.

Version 2 — Defined Audience

Layer Added

Defined audience

Prompt

Help me frame a machine-learning approach for improving the existing search-ranking pipeline. I want to understand the problem and decide what should be improved before choosing a complex model. Explain it for an ML intern who

already understands basic machine learning but is working on a real search-ranking task.

Output — Representative Excerpt

Start by defining the ranking objective and identifying the current baseline. Since you already understand basic ML concepts, focus on how the ranking problem maps to features, predictions and evaluation rather than explaining basic concepts such as what supervised learning is.

What Changed?

I specified that the answer is for an **ML intern with basic ML knowledge**.

What Improved?

The response stopped spending as much space on basic ML explanations and became more relevant to the actual work.

What Still Failed?

It still lacked the **specific FlyRank context and observed baseline/model gap**.

What I Would Try Next

Add **real context**.

Version 3 — Real Context

Layer Added

Real context

Prompt

Help me frame a machine-learning approach for improving the existing search-ranking pipeline. I want to understand the problem and decide what should be improved before choosing a complex model. Explain it for an ML intern who already understands basic machine learning but is working on a real search-ranking task. This is part of my FlyRank internship. In the initial discovery work, the baseline performance was around 0.24 while the reference model was around 0.74.

Output — Representative Excerpt

The large gap between the baseline (~0.24) and reference model (~0.74) suggests that the first priority should be understanding why the existing pipeline performs poorly. Compare the data, features, ranking logic and evaluation process before attempting additional model complexity.

What Changed?

I added the **actual FlyRank context and baseline/reference performance**.

What Improved?

The response became much more specific. Instead of giving general advice, it focused on the **0.24 → 0.74 performance gap** and suggested investigating its source.

What Still Failed?

The answer was useful but not organized in a way that I could directly use for my internship notebook.

What I Would Try Next

Add a **specified output format**.

Version 4 — Specified Output Format

Layer Added

Specified output format

Prompt

Help me frame a machine-learning approach for improving the existing search-ranking pipeline. I want to understand the problem and decide what should be improved before choosing a complex model. Explain it for an ML intern who already understands basic machine learning but is working on a real search-ranking task. This is part of my FlyRank internship. In the initial discovery work, the baseline performance was around 0.24 while the reference model was around 0.74.

Structure the response using four headings: Problem, Evidence, What to Investigate, and Next Step.

Output — Representative Excerpt

Problem

The current ranking pipeline has a substantial performance gap compared with the reference model.

Evidence

The baseline is approximately 0.24 compared with approximately 0.74 for the reference model.

What to Investigate

Examine the data, features, ranking process and evaluation before adding model complexity.

Next Step

Build or inspect a readable baseline and use it to identify where improvement is possible.

What Changed?

I added only a **four-section output structure**.

What Improved?

The answer became easier to use directly in my internship work because the important information was separated into clear sections.

What Still Failed?

The structure improved usability, but it could still produce confident recommendations without clearly distinguishing **known information from assumptions**.

What I Would Try Next

Add a **verification requirement**.

Version 5 — Verification Requirement

Layer Added

Verification requirement

Final Prompt

Help me frame a machine-learning approach for improving the existing search-ranking pipeline. I want to understand the problem and decide what should be improved before choosing a complex model. Explain it for an ML intern who already understands basic machine learning but is working on a real search-ranking task. This is part of my FlyRank internship. In the initial discovery work, the baseline performance was around 0.24 while the reference model was around 0.74.

Structure the response using four headings: Problem, Evidence, What to Investigate, and Next Step.

Clearly separate facts provided in the context from assumptions or recommendations. Do not invent metrics, dataset details or model results. If something is unknown, explicitly say that it is unknown.

Output — Representative Excerpt

Problem

The known problem is a gap between the existing baseline performance (~0.24) and the reference model (~0.74).

Evidence

These performance values come from the initial discovery work. No additional dataset or feature information has been provided.

What to Investigate

The next investigation should focus on the data, features, ranking process and evaluation metric rather than assuming that a more complex model is required.

Next Step

Inspect the existing workflow and establish a readable baseline before making further model changes.

What Changed?

I added only a **verification requirement** telling AI not to invent information and to separate facts from assumptions.

What Improved?

The output became more trustworthy. It no longer treated missing project details as facts and clearly separated what I actually knew from what should be investigated.

What Still Failed?

This did not dramatically change the technical recommendation. The main improvement was **reliability rather than new information**.

This was useful because it showed that not every prompt change produces a dramatic improvement in the answer.

What I Would Try Next

For future prompts, I would add more specific project information only when I have verified it, rather than adding unnecessary instructions.

Final Reusable Prompt

Help me frame a machine-learning approach for improving the existing search-ranking pipeline.

I am an ML intern working on Search Ranking Improvement at FlyRank. I already understand basic machine learning, so focus on the actual ranking problem rather than explaining basic ML concepts.

In the initial discovery work, the baseline performance was approximately 0.24 while the reference model was approximately 0.74. I want to understand the source of this gap before choosing a more complex model.

Structure the response using:

1. **Problem** — define the ML problem clearly.

2. **Evidence** — use only the information I provide.
3. **What to Investigate** — identify the most important things to examine before changing the model.
4. **Next Step** — give one practical next action.

Do not invent dataset details, metrics, features or model results. Clearly distinguish known facts from assumptions. Prefer a simple, readable approach before recommending additional complexity.

Conclusion

The ladder showed that prompt quality improved when I added information **one layer at a time**.

The biggest improvement came from adding **real context**, because the response changed from generic machine-learning advice to advice connected to the actual FlyRank ranking problem. The output format then made the response easier to use, while the verification requirement made it safer and more trustworthy.

The main lesson was that a longer prompt is not automatically a better prompt. Each added instruction should have a specific purpose and should improve something observable in the output.

Claude Project Configuration / Framing a machine-learning search-ranking improvement

Problem

The search-ranking pipeline's current model underperforms relative to a reference model (a large gap in whatever metric is being tracked). The task is to identify why this gap exists before jumping to a more complex model, since added complexity without understanding the failure modes rarely closes a performance gap efficiently.

Evidence

Fact (given): Baseline performance was around 0.24, and the reference model's performance was around 0.74.
Unknown: The specific metric being measured (e.g., NDCG, MRR, MAP, precision@k) is not stated.
Unknown: What "reference model" means here — whether it is a stronger baseline, a production system, or an upper-bound benchmark — is not specified.
Unknown: Details of the dataset, features, query volume, or label quality are not provided.

What to Investigate

- Clarify the exact metric and how it's computed, since a 0.24 vs 0.74 gap means very different things depending on the metric. ↓
- Check data and label quality: are relevance labels reliable, consistent, and representative

- **Examine feature coverage:** Does the current pipeline use meaningful ranking signals (query-document relevance, click behavior, freshness, popularity), or is it relying on very few/weak features?
- **Analyze error patterns:** Are failures concentrated in certain query types (e.g., rare/long-tail queries), or spread evenly?
- **Compare the reference model's approach:** Is its advantage coming from better features, more data, or a fundamentally different algorithm?
- **Assess pipeline mechanics:** Check for bugs in feature computation, data leakage, or mismatched train/test distributions before assuming the model itself is the bottleneck.

Next Step

Recommendation (not a fact): Before selecting a more complex model, perform a diagnostic analysis — audit data quality, feature set, and error patterns — to determine whether the performance gap is caused by data/feature limitations or by genuine model capacity limitations. This diagnosis should guide whether the next step is feature engineering, data improvement, or model architecture changes.

